but the constraint was not defined that way initially, you'll need to drop the constraint and create it again using that option.

Next, you'll try out a less common but powerful constraint type called the Exclusion (EXCLUDE) Constraint. The big difference with this type is that it's used to constrain data across multiple rows. How does that work?

Preventing Overlaps With An Exclusion Constraint

Instead of a condition to enforce for a single row, an Exclusion constraint works at the level of multiple rows and can be used to express conditions about when they shouldn't match or overlap.

In Rideshare, Vehicle Reservations should not overlap. Two Riders should not be able to reserve the same vehicle at the same time. Typically this business logic will be written in the application. However this is mission critical business logic and you've decided you'd like to enforce it within PostgreSQL.

Overlapping Vehicle Reservations would cause a very bad experience for each person involved. Not only would the reservation be impossible to fulfill for both people simultaneously, it would be really unclear which reservation to honor. The first one? The most recent one? Someone is going to be upset.

That's something you'd definitely want to avoid.

Take a moment to think through how to define this constraint. Only one Vehicle should be reserved by a Rider at a time. You'll use an Exclusion constraint (EXCLUDE) to detect when an overlap would occur and the constraint will prevent the overlap.

To create the Exclusion constraint you'll use a GiST index. You haven't yet worked with GiST indexes; for now it's enough to know that they are the Index type you need for your Exclusion constraint.

The SQL statement to add this constraint type is listed below. This Exclusion constraint has already been added to Rideshare, so you may explore the definition by running \d vehicle_reservations from psql. Each of the pieces will be explained in the next section.

sql/exclusion_constraint.sql

```
ALTER TABLE vehicle_reservations
DROP CONSTRAINT IF EXISTS non_overlapping_vehicle_registration;

CREATE EXTENSION IF NOT EXISTS btree_gist;

ALTER TABLE vehicle_reservations
ADD CONSTRAINT non_overlapping_vehicle_registration
EXCLUDE USING gist (
```